UNSUPERVISED LEARNING

PROJECT ON -

‘CREDIT CARD CUSTOMER SEGMENTATION’

BUSINESS REPORT

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Problem Statement

Context

AllLife Bank wants to focus on its credit card customer base in the next financial year. They have been advised by their marketing research team, that the penetration in the market can be improved. Based on this input, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers. Another insight from the market research was that the customers perceive the support services of the back poorly. Based on this, the Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster. The Head of Marketing and Head of Delivery both decide to reach out to the Data Science team for help.

Objective

To identify different segments in the existing customers, based on their spending patterns as well as past interaction with the bank, using clustering algorithms, and provide recommendations to the bank on how to better market to and service these customers.

Data Description

The data provided is of various customers of a bank and their financial attributes like credit limit, the total number of credit cards the customer has, and different channels through which customers have contacted the bank for any queries (including visiting the bank, online, and through a call center).

Data Dictionary

- SI_No: Primary key of the records
- Customer Key: Customer identification number

- Average Credit Limit: Average credit limit of each customer for all credit cards
- Total credit cards: Total number of credit cards possessed by the customer
- Total visits bank: Total number of visits that the customer made (yearly) personally to the bank
- Total visits online: Total number of visits or online logins made by the customer (yearly)
- Total calls made: Total number of calls made by the customer to the bank or its customer service department (yearly)

Data Overview

Shape of the dataset

- Dataset has 660 rows and 7 columns.

View first and last five rows

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1	87073	100000	2	1	1	0
1	2	38414	50000	3	0	10	9
2	3	17341	50000	7	1	3	4
3	4	40496	30000	5	1	1	4
4	5	47437	100000	6	0	12	3
...	...	...	...	...	...	...	...
655	656	51108	99000	10	1	10	0
656	657	60732	84000	10	1	13	2
657	658	53834	145000	8	1	9	1
658	659	80655	172000	10	1	15	0
659	660	80150	167000	9	0	12	2

660 rows × 7 columns

Table 1: First & Last 5 Rows

Checking the Data Types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660 entries, 0 to 659
Data columns (total 7 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Sl_No                660 non-null    int64
 1   Customer Key         660 non-null    int64
 2   Avg_Credit_Limit     660 non-null    int64
 3   Total_Credit_Cards   660 non-null    int64
 4   Total_visits_bank    660 non-null    int64
 5   Total_visits_online  660 non-null    int64
 6   Total_calls_made     660 non-null    int64
dtypes: int64(7)
memory usage: 36.2 KB
```

Table 2: Data Types

- All the seven columns in the dataset are integer variables.
- All the columns have non-null count equal to total number of entries (rows) in the dataset. So, dataset has no missing values.
- ## Checking for Unique Values

	0
Sl_No	660
Customer Key	655
Avg_Credit_Limit	110
Total_Credit_Cards	10
Total_visits_bank	6
Total_visits_online	16
Total_calls_made	11

dtype: int64

Table 3: Unique Values

- Sl_No has all the values unique (i.e. 660, same as total number of rows in the dataset). Also, Customer Key has almost all the values unique (i.e. 655).
- Avg_Credit_Limit has very high cardinality i.e. 110 unique values.
- Total_Credit_Cards and Total_visits_bank have low cardinality i.e. 10 and 6 unique values respectively.
- Total_visits_online and Total_calls_made have moderate to high cardinality i.e. 16 and 11 unique values respectively.

Checking for Duplicate Values

- **Dataset has no duplicate values.**

Checking for Missing Values

	0
Sl_No	0
Customer Key	0
Avg_Credit_Limit	0
Total_Credit_Cards	0
Total_visits_bank	0
Total_visits_online	0
Total_calls_made	0

dtype: int64

Table 4: Missing Values

- **Dataset has no missing values.**

Statistical Summary

Five-Point Summary

	count	mean	std	min	25%	50%	75%	max
Sl_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.443939	25627.772200	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.00	18000.0	48000.00	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Table 5: Five-Point Summary

- Avg_Credit_Limit varies from 3K to 2Lacs with mean around 34.5K , median 18K and std dev ~38K. This shows data is right skewed . 75% values are less than 48K

- Total credit cards possessed by the customer range from 1 to 10 with 23% people having 4 , 18% 6 , 15% 7 , 11% 5 , 10% 2 , 9 % 1 , 8% 3 , 3 % 10 and 2% have 8 and 9 cards respectively.
- 24% customers have visited the bank twice, 17% once and 15% 0 , 3 and 5 times and 14% 4 times in an year
- Online visits range from 0 to 15 with 75% customers visiting less than 4 times a year
- Total number of calls made by the customer to the bank or its customer service department (yearly) range from 0 to 10 with 75% customers contacting less than 5 times a year.

Remove Insignificant Columns

As we have seen that Sl_No has all the values unique (i.e. 660, same as total number of rows in the dataset). Also, Customer Key has almost all the values unique (i.e. 655). So, these columns are insignificant and irrelevent, as they are not adding any value to the analysis and model/cluster building. Thus, we are dropping these columns.

Exploratory Data Analysis

Univariate Analysis

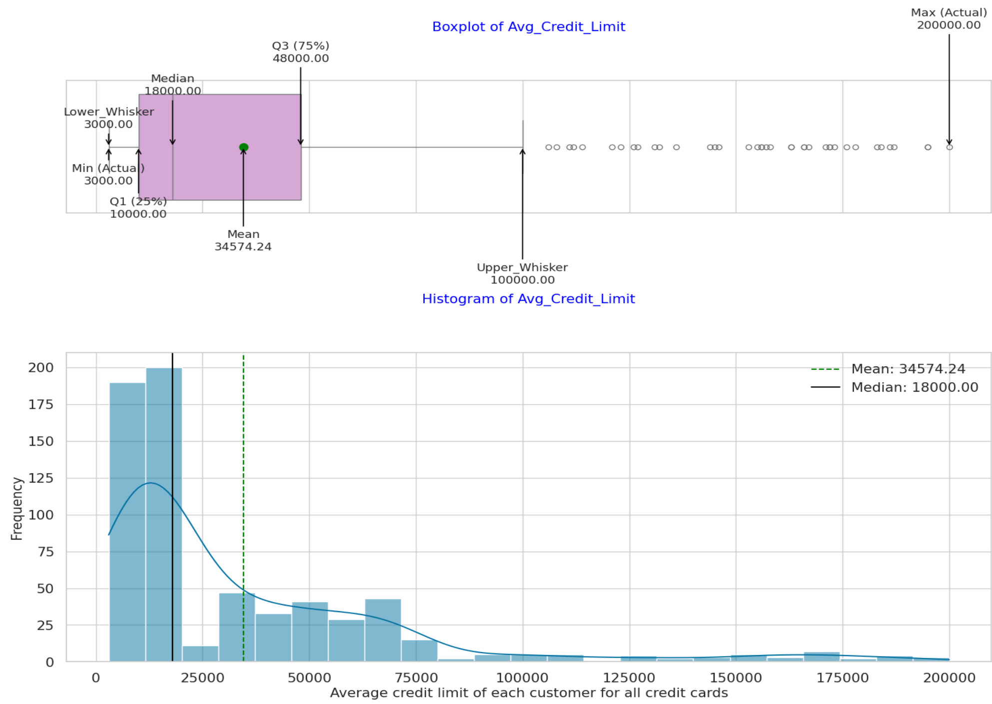

Figure 1: Combined Boxplot & Histogram of Average_ Credit_ Limit

- Data is highly right skewed with mean at ~35K and median at ~18K.
- Values range from 3K to 2 Lacs, thus variability is very high.
- 75% customers have average credit limit equal to and below 50K.
- There are several outliers, especially in high range (We won't be treating them as they are genuine values).

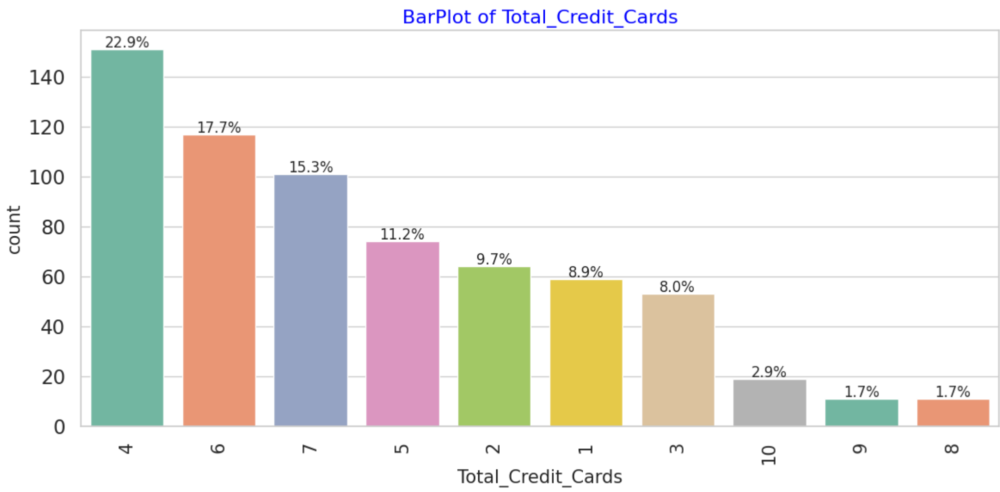

Figure 2: Labeled Barplot of Total_Credit_cards

- Majority people have 4 credit cards (23%), followed by 5-7 credit cards (44%), followed by 1-3 credit cards (27%) and trace for 8-10 credit cards (6%).

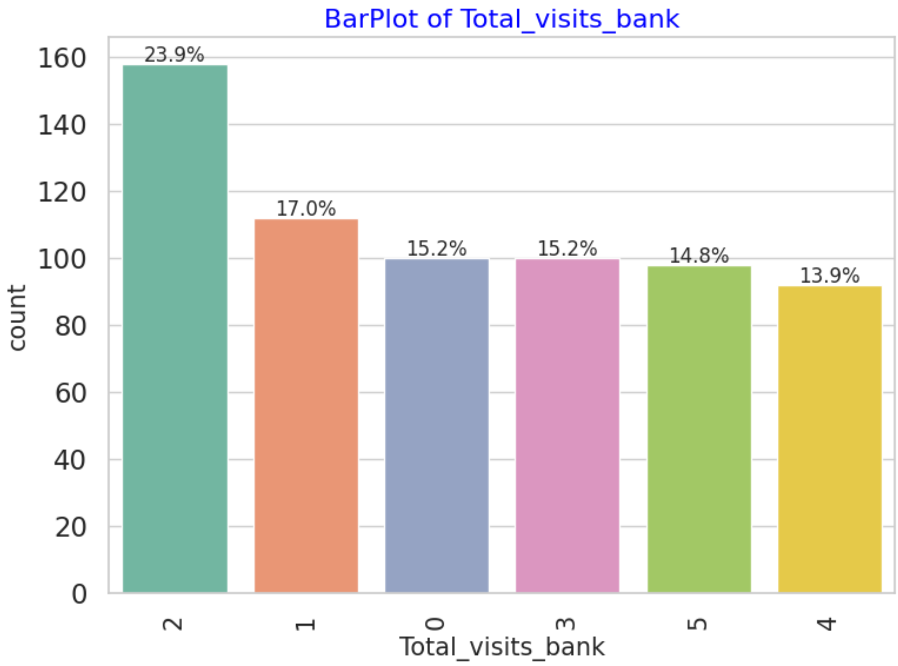

Figure 3: Labeled Barplot of Total_visits_bank

- Most customers visited the bank 2 times (approx. 24%) indicating that bi-annually visits may be the most common behavior.
- About 17% visited only once, and 15.2% didn't visit at all. This suggests a sizable group is either new, inactive, or prefers digital banking.
- Around 15.2% visited 3 times, and 14.8% visited 5 times. Customers visiting 4 times are the least common (13.9%). This shows that frequent physical branch users are still a consistent, though not dominant segment.
- Digital engagement opportunities: The 15%+ customers with 0 or 1 visit could be nudged towards digital services or targeted for re-engagement campaigns.
- High-engagement customers (3–5 visits) might be better served with loyalty programs, priority services, or personal banking offers.
- Consider optimizing branch operations around peak footfall frequencies (like 2 visits/year segments).

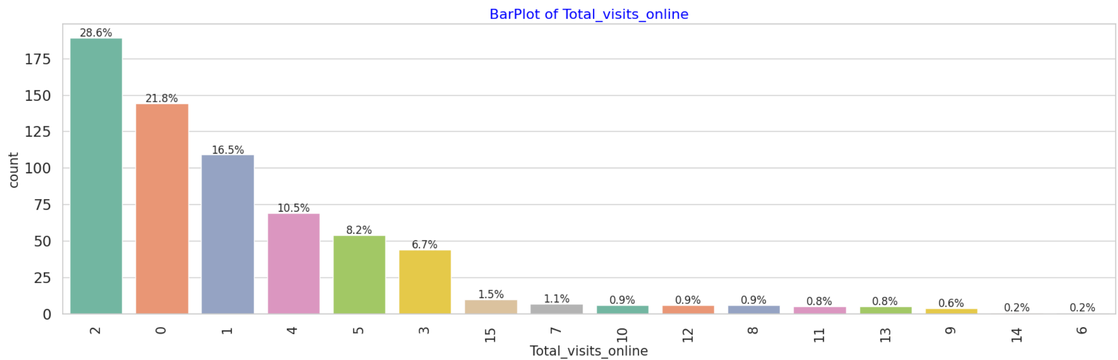

Figure 4: Labeled Barplot of Total_visits_online

- Total online visits are max at 0-2 per year (67% customers), showing less digital orientation and might be the preference for physical banking. There is Digital Adoption Opportunity to focus on converting these users through digital onboarding, app training, and incentives.
- Almost 25% people have 3-5 online visits per year, indicating decent digital presence.
- Remaining traces (around 8% people) visit online more than 5 times annually. These are the tech-savvy people who might need better digital services, online offers, etc.

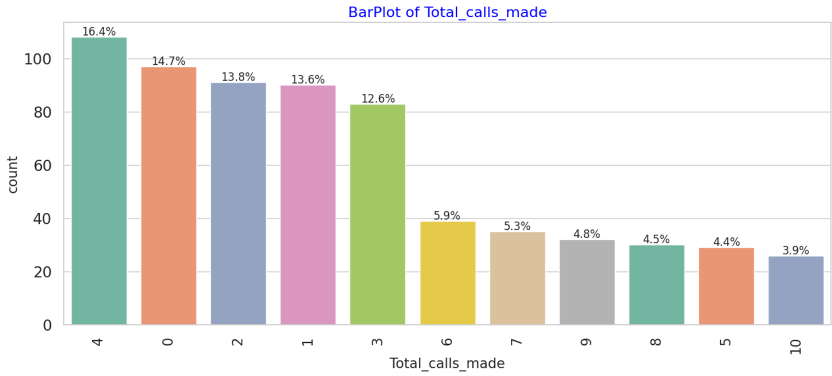

Figure 5: Labeled Barplot of total_calls_made

- Majority customers lie in 0-4 calls/year (71%), indicating either low or no engagement. Company should see if the issues are less and get resolved faster or the customer base is not having faith in the services
- 29% people who make 5 or more calls annually, especially 13% people who make 8-10 calls need attention. They might be facing repetitive issues and need attention to retain them.

Bivariate Analysis

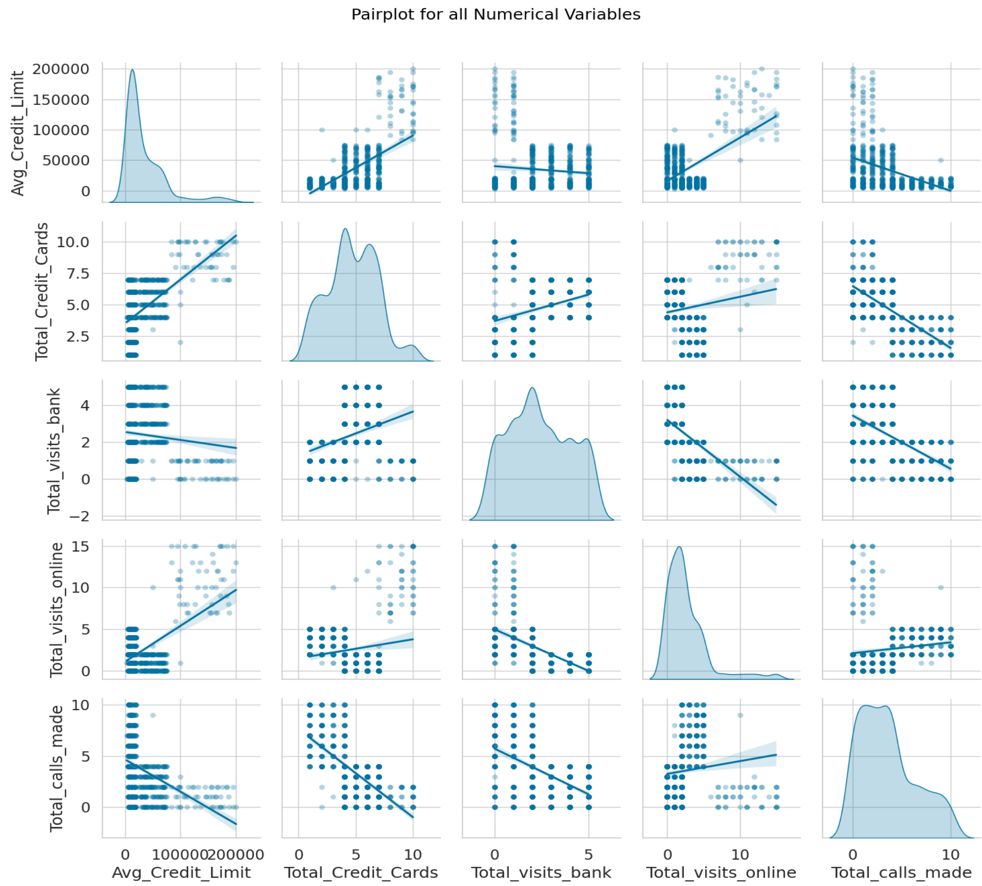

Figure 6: Pairplot

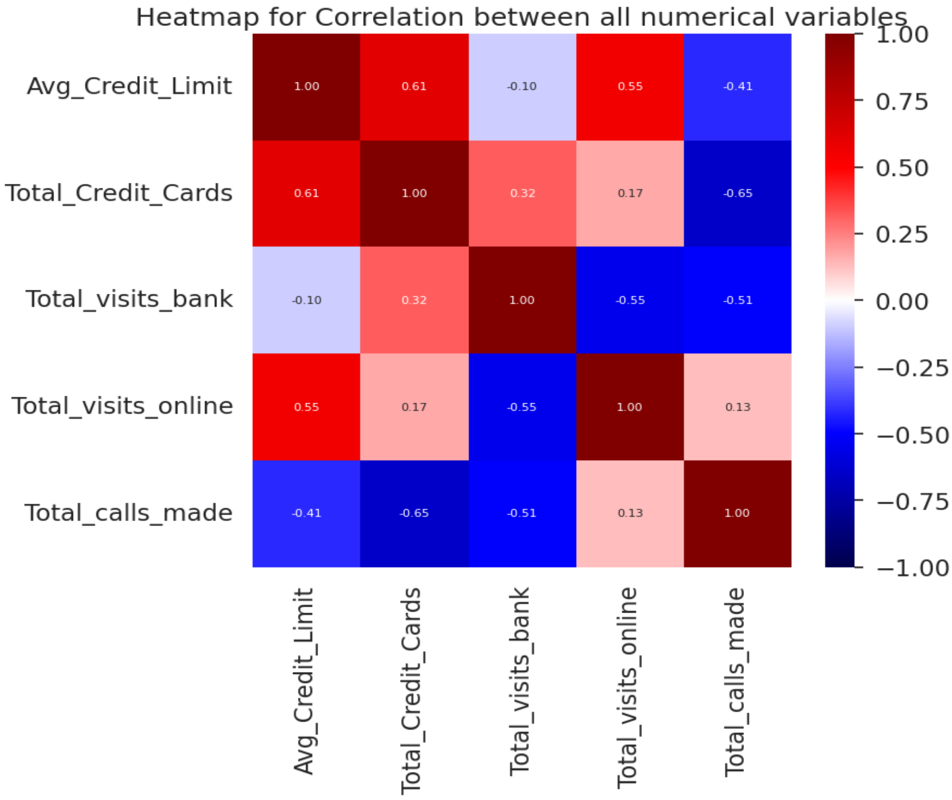

Figure 7: Correlation Heatmap

- Customers with More credit cards have higher Avg Credit Limit, visit bank more and have less phone calls.
- Customers with higher Avg Credit Limit tend to visit online more and make less phone calls, thus digitalisation is preferred here.
- Customers who visit bank more, visit online less and make less use of phone banking. These are traditional and less digitally aligned customers that prefer in person banking.
- Customers who make more phone calls, have less average credit limit, credit cards and make less bank visits. This is mostly dissatisfied customer base that needs to be tapped by resolving their issues, digital training , etc.

Data Pre-Processing

- Insignificant columns Sl_No and Customer Key have already been dropped as they add no value to the analysis and so to model/cluster building.

- We have already checked for duplicate values, missing values and we have not witnessed any of them. Also, we have not observed any anamoly, irregularity till yet. So, the data is clean to proceed furhter in the analysis.

Outlier Detection

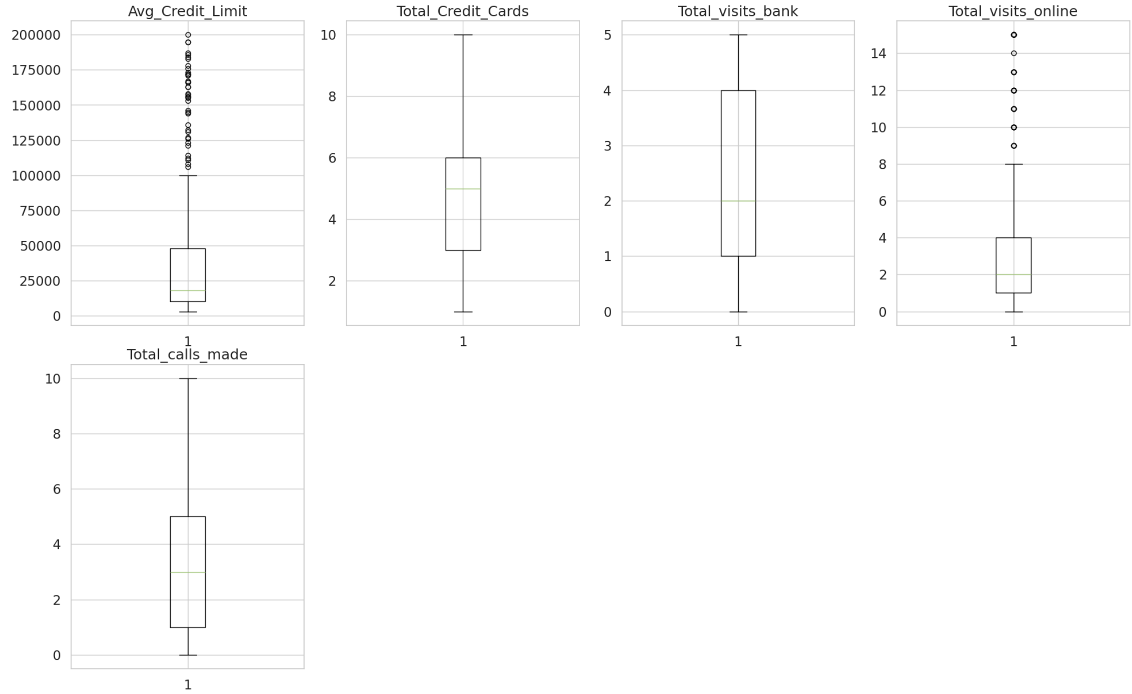

Figure 8: Boxplots for Outlier Detection

- There are several outliers present in Avg_Credit_Limit and Total_visits_online, especially in higher ranges.
- **Outliers are NOT being treated as these are real values and represent real world data.**

Data Scaling

Let's scale the data before we proceed further with clustering because clustering algorithms, particularly distance-based ones like K-Means and hierarchical clustering, are sensitive to the scale of the features. Features with larger values can dominate the distance calculations, leading to biased results.

StandardScaler applies Z-score standardization, transforming each feature to have a mean of 0 and a standard deviation of 1. This brings all features to a similar scale without distorting the differences in the ranges of values, which is important for accurately calculating distances between data points and thus forming meaningful clusters based on all features equally.

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1.740187	-1.249225	-0.860451	-0.547490	-1.251537
1	0.410293	-0.787585	-1.473731	2.520519	1.891859
2	0.410293	1.058973	-0.860451	0.134290	0.145528
3	-0.121665	0.135694	-0.860451	-0.547490	0.145528
4	1.740187	0.597334	-1.473731	3.202298	-0.203739

Table 6: First Five Rows of Scaled Data

Model Building - K-means Clustering

Checking Elbow Plot

Number of Clusters	Average Distortion
2	1.71
3	1.14
4	1.09
5	0.99
6	0.95
7	0.90
8	0.91
9	0.89
10	0.83

Table 7: Number of Rows & Avg Distortion

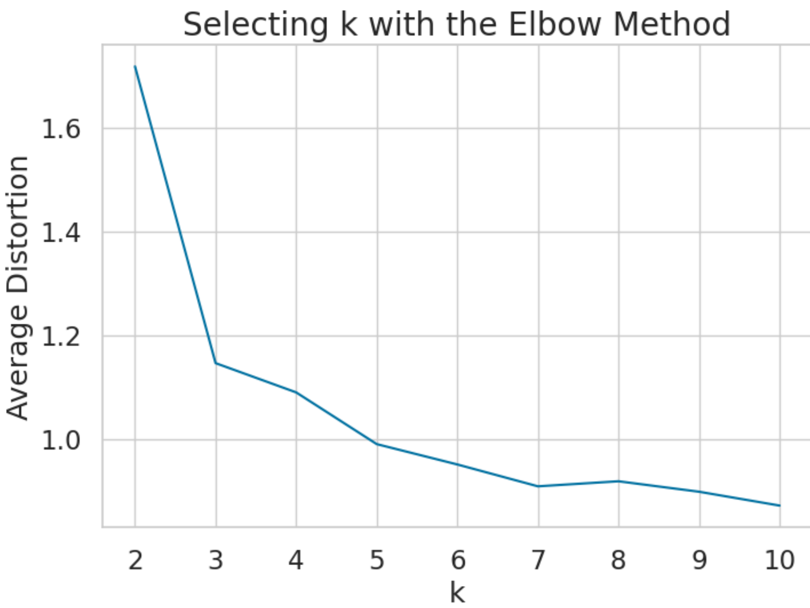

Figure 9: Plot of Elbow Method

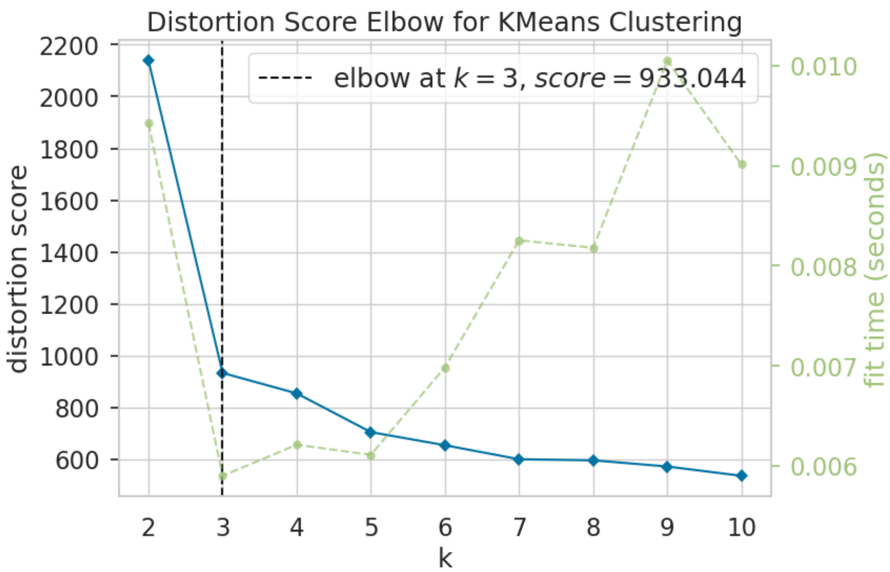

Figure 10: Plot of Elbow Method with Optimal 'k'

- From the Elbow Plot (Distortion Score), it seems 3 is good value for k.

Checking Silhouette Score

Number of Clusters	Silhouette Score
2	0.570
3	0.515
4	0.374
5	0.271
6	0.248
7	0.247
8	0.225
9	0.199
10	0.209

Table 8: Number of Clusters with Silhouette Score

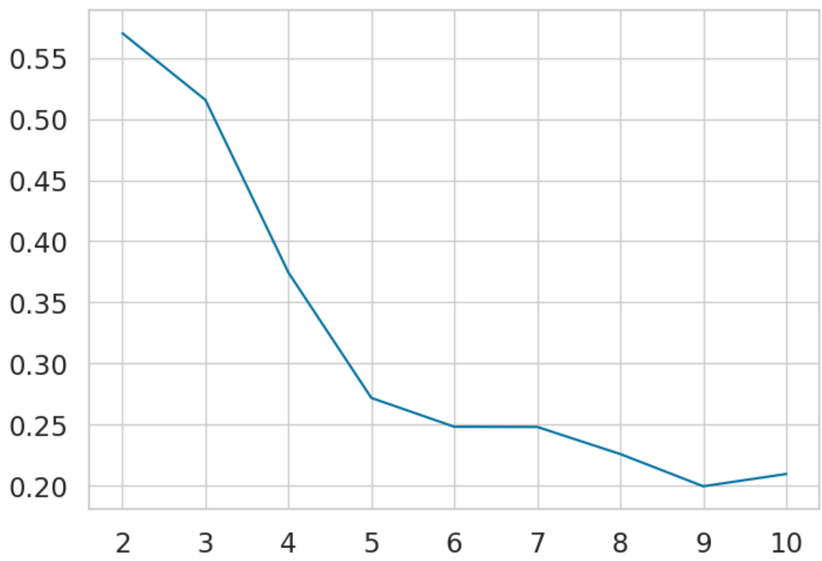

Figure 11: Silhouette Score vs Number of Clusters

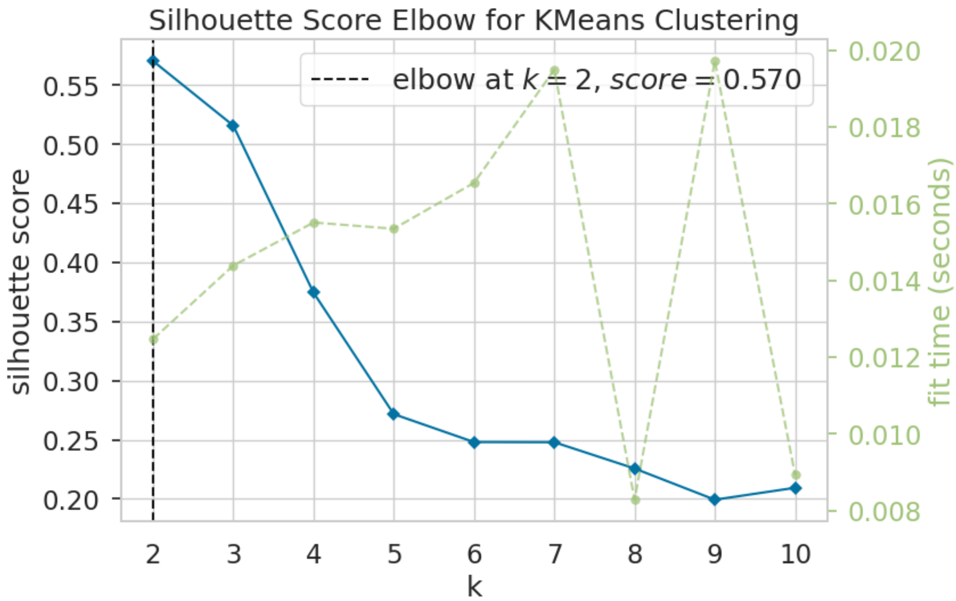

Figure 12: Silhouette Score vs Number of Clusters with Optimal 'k'

- From the Silhouette Score, it seems that 2 is the good value for k.

Silhouette Plot

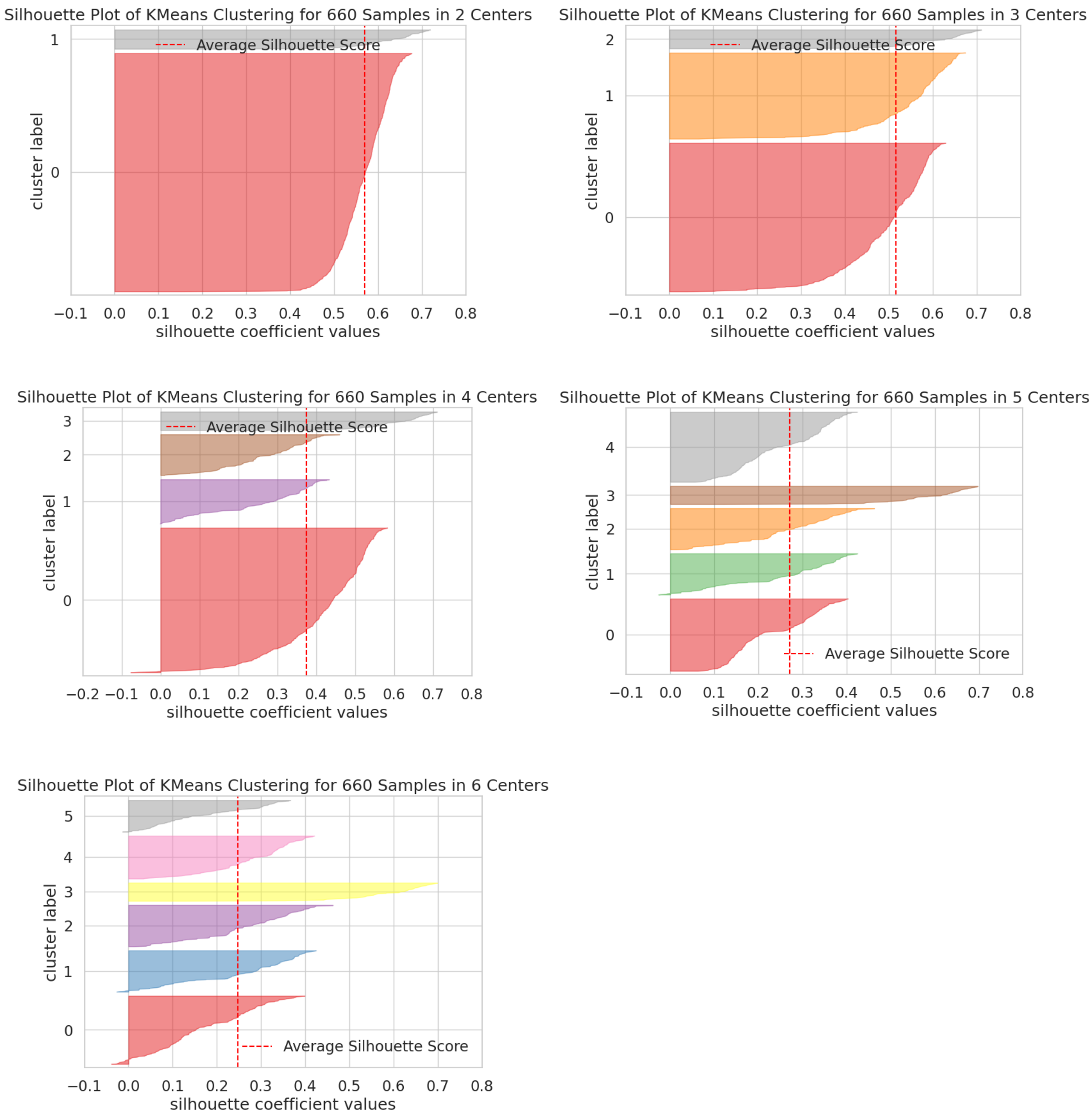

Figure 13: Cluster Label vs Silhouette Coefficient Values

We will proceed with k=3 (optimal number of clusters) because

- **The Average Silhouette Score is around 0.515, which is good (above 0.5).**
- **Though k=2 have little bit better Average Silhouette Score 0.570 than that of k=3, but the elbow in the Elbow Plot (Distortion Score) for k=3 is reasonable than that for k=2.**
- **Clusters are relatively well-separated, with NO negative Silhouette scores and thus no misclassified points.**

Cluster Profiling

K_means_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segment
0	33782.383420	5.515544	3.489637	0.981865	2.000000	386
1	12174.107143	2.410714	0.933036	3.553571	6.870536	224
2	141040.000000	8.740000	0.600000	10.900000	1.080000	50

Table 9: K_means _segments Cluster Profile

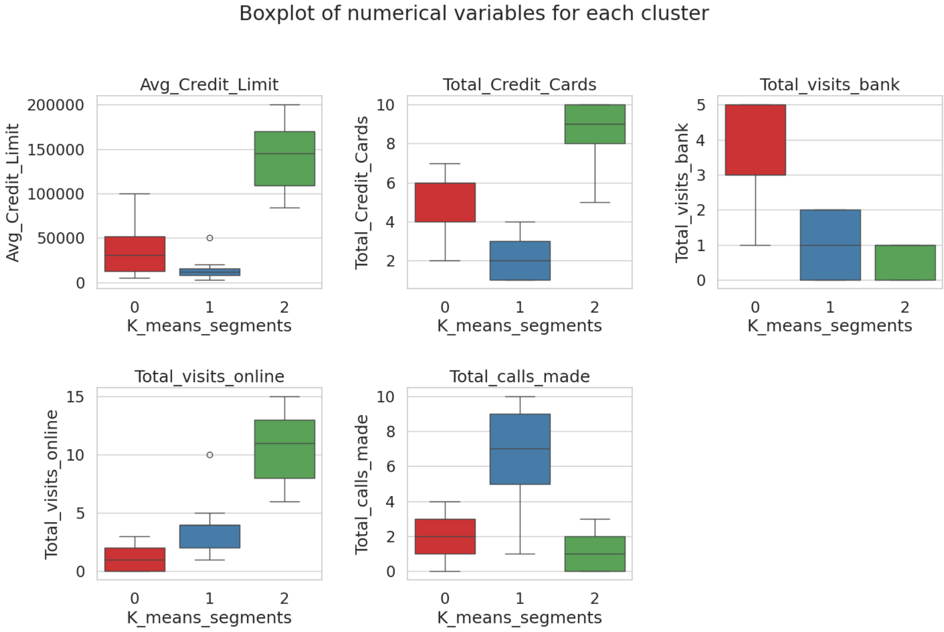

Figure 14: Boxplot of Original Numerical Variables for Each Cluster

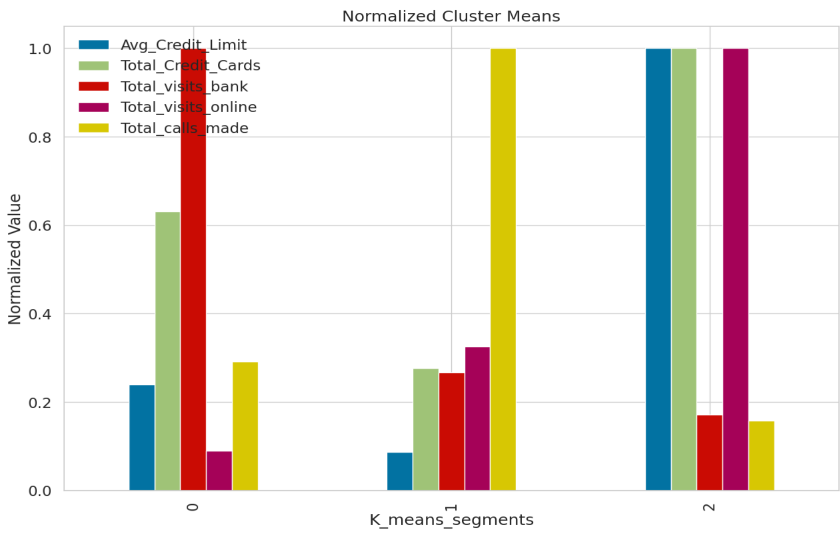

Figure 15: Normalised Mean Values vs K_means_segments

Insights

Cluster 0:

- Highest Total_visits_bank
- High Total_Credit_Cards
- Low Total_visits_online
- Likely traditional customers who frequently visit banks physically and have multiple credit cards due to years of banking relationship , these are possibly old customers with traditional banking.

Cluster 1:

- High Total_calls_made
- Low Avg_Credit_Limit
- Moderate values for other features
- Possibly represents high-touch, lower credit customers who rely more on phone calls than digital banking.

Cluster 2:

- Highest Avg_Credit_Limit, Total_Credit_Cards, and Total_visits_online
- Low Total_visits_bank and Total_calls_made
  Probably digitally-savvy premium customers with high credit limits and less reliance on traditional or phone channels. Bank should focus more on these with premium offering and features as they are high profit and tech savvy customers who are crucial to maintain and need more focus , otherwise they may migrate to competitors if not satisfied by the services.

Model Building - Hierarchical Clustering

Computing Cophenetic Correlation

Distance Metric	**Linkage Method**	**Cophenetic Correlation**
Euclidean	Single	0.7391
Euclidean	Complete	0.8599
Euclidean	Average	0.8977
Euclidean	Weighted	0.8861
Chebyshev	Single	0.7382
Chebyshev	Complete	0.8533
Chebyshev	Average	0.8974
Chebyshev	Weighted	0.8913
Mahalanobis	Single	0.7058
Mahalanobis	Complete	0.6663
Mahalanobis	Average	0.8326
Mahalanobis	Weighted	0.7805
Cityblock	Single	0.7252
Cityblock	Complete	0.8731
Cityblock	Average	0.8963
Cityblock	Weighted	0.8825

Table 10: Cophenetic Correlation

Exploring different linkage methods with Euclidean distance only.

Linkage Method	Cophenetic Correlation
Single	0.739
Complete	0.859
Average	0.897
Centroid	0.893
Ward	0.741
Weighted	0.886

Table 11: Cophenetic Correlation with Euclidean Distance

- **We see that cophenetic correlation is maximum (0.897) with Euclidean distance and Average linkage.**

Checking Dendrograms

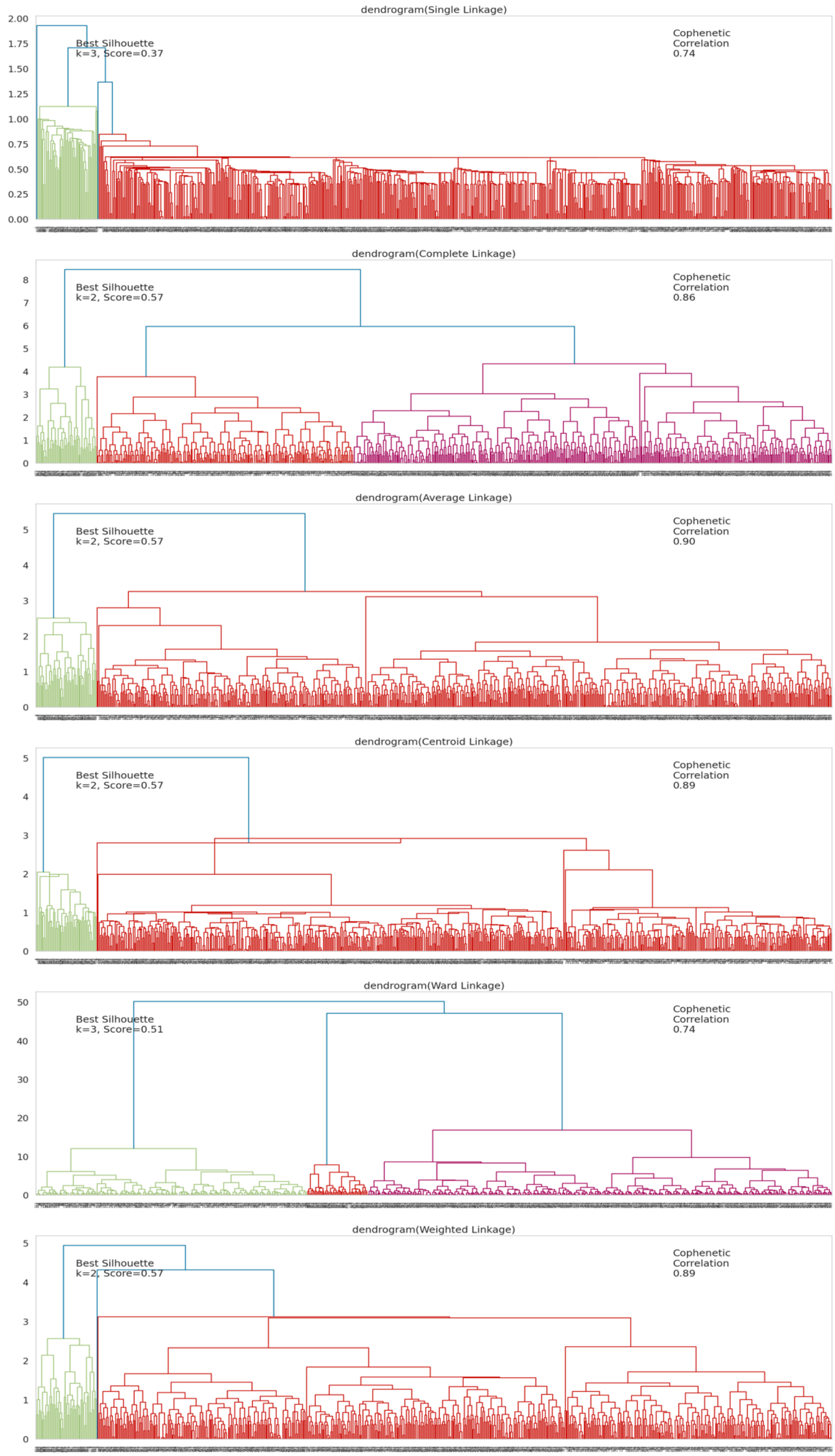

Figure 16: Dendrograms

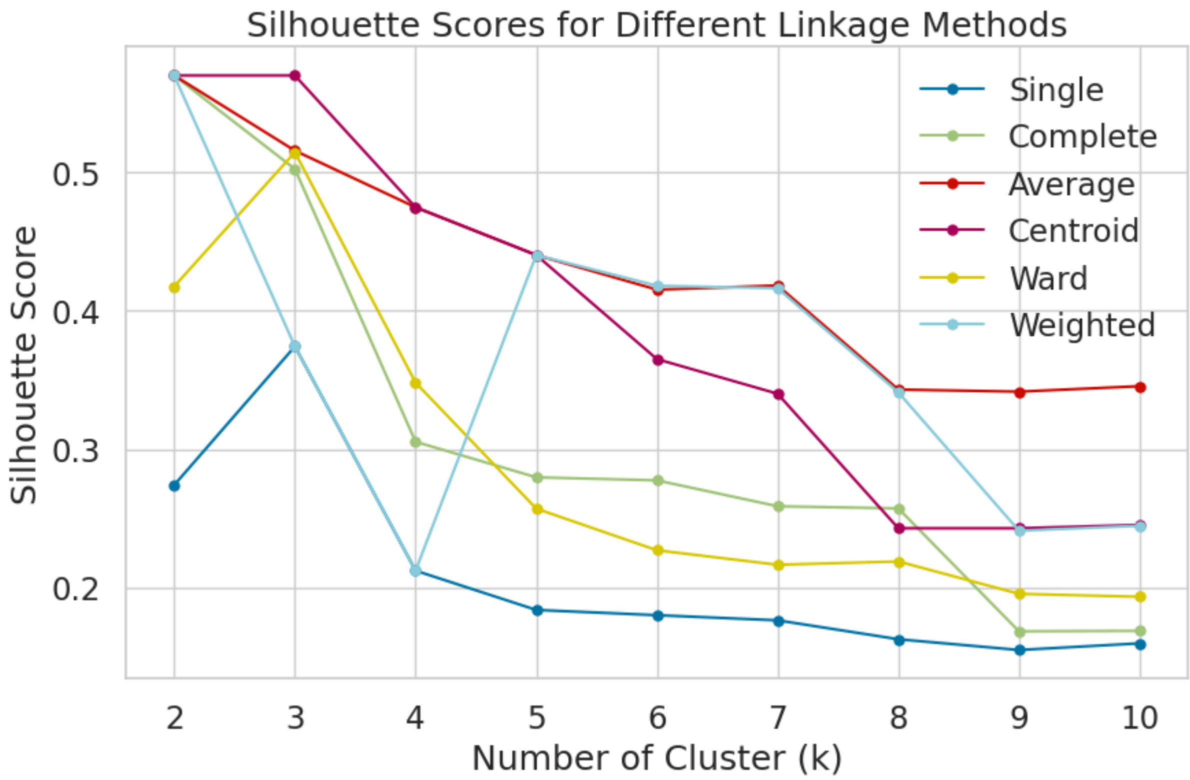

Figure 17: Lineplot for Silhouette Scores for Different Linkage Methods

```
Silhouette Score Summary:
Single Linkage: Best k = 3, Silhouette Score = 0.374
Complete Linkage: Best k = 2, Silhouette Score = 0.570
Average Linkage: Best k = 2, Silhouette Score = 0.570
Centroid Linkage: Best k = 2, Silhouette Score = 0.570
Ward Linkage: Best k = 3, Silhouette Score = 0.515
Weighted Linkage: Best k = 2, Silhouette Score = 0.570
CPU times: user 25.3 s, sys: 212 ms, total: 25.5 s
Wall time: 31.4 s
```

Table 12: Silhouette Score Summary

- **The cophenetic correlation is highest for average and centroid linkage methods.**
- **We will move ahead with average linkage.**
- To determine the optimal number of clusters from the dendrogram we apply the horizontal cut method:

a. Draw a horizontal line across the dendrogram where the vertical distance (height) between clusters is largest before merges — this indicates well-separated clusters.
b. Count the number of vertical lines intersected by your horizontal cut — that's optimal number of clusters.

c. The biggest vertical jump is near the top (height ~5.5), where the last two clusters are merged into one (blue line).
d.The tallest vertical line is the one merging the 2 main groups, around height = 5.5.
e. The second tallest jumps occur around height = 3.3.
f. A horizontal cut at ~3.3 intersects 3 main vertical branches → suggesting 3 clusters.
g. A little below that , around ~3.1, we get 4 clusters but there is a single customer being segmented out to a new cluster, which is of no use

- Thus, 3 appears to be the appropriate number of clusters from the dendrogram for average linkage because it gives

1. A balance between granularity and cohesion.
2. 3 clusters that are well-separated before a major merge happens.
3. Cutting higher gives only 2 very broad clusters (too coarse).
4. Cutting lower gives 4–5 clusters, but they are increasingly similar, possibly over-segmenting the data.

- **However, since Silhouette score is better at k = 2 , we will first try k=2 and see if it makes business sense**

Cluster Profiling (k=2)

HC_Clusters	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segment
0	25847.540984	4.375410	2.550820	1.926230	3.788525	610
1	141040.000000	8.740000	0.600000	10.900000	1.080000	50

Table 13: HC_Clusters Profile with k=2

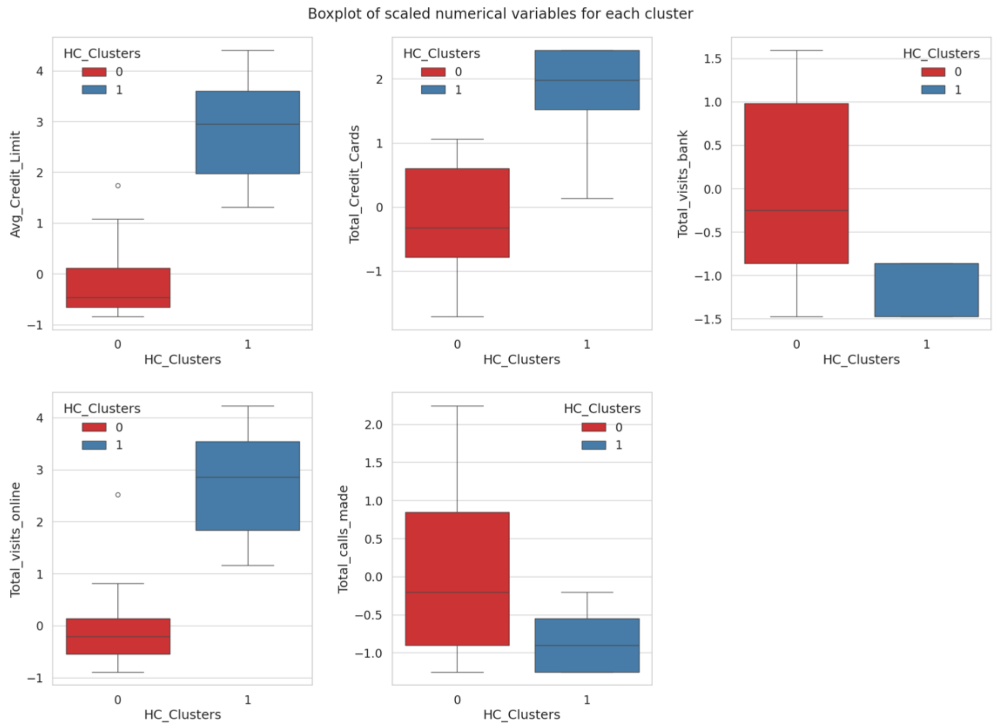

Figure 18: Boxplot of scaled numerical variables for each cluster (Hierarchical Clustering & k=2)

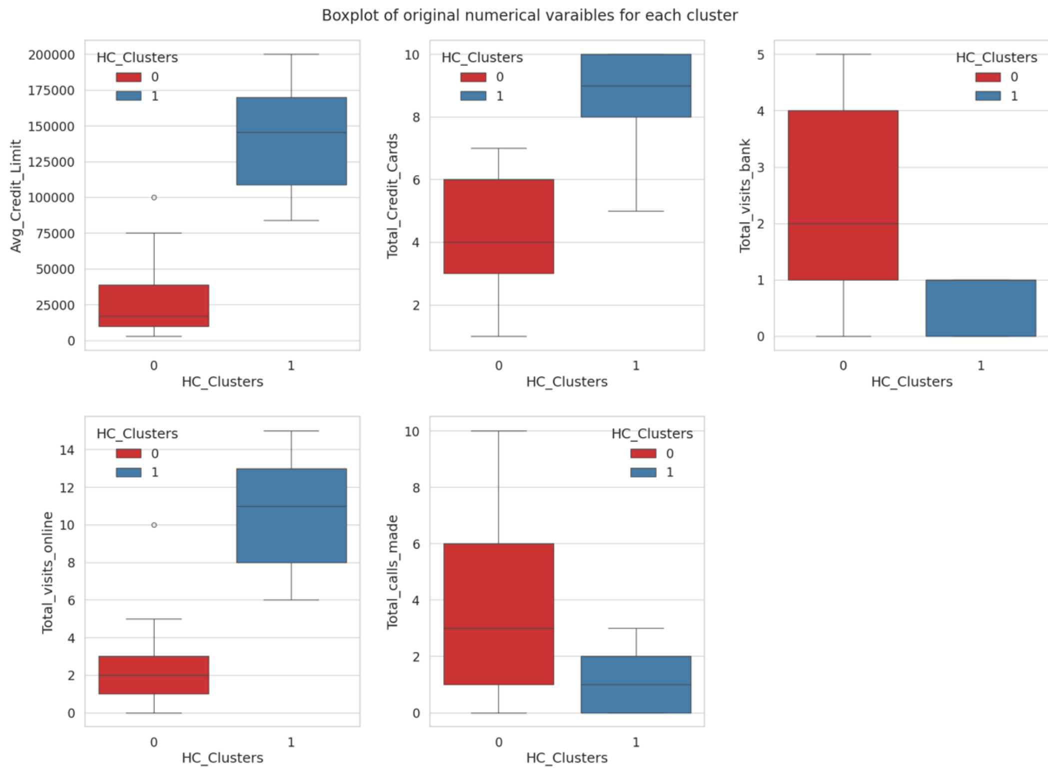

Figure 19: Boxplot of original numerical variables for each cluster (Hierarchical Clustering & k=2)

Insights

Cluster 0 :

- Avg_Credit_Limit: ₹25,847 → significantly lower than Cluster 1.
- Total_Credit_Cards: 4.38 cards → moderate usage of credit products.
- Total_visits_bank: 2.55 → prefers offline or in-person interactions.
- Total_visits_online: 1.93 → low online engagement.
- Total_calls_made: 3.79 → relatively high phone interaction.
- Count: 610 customers → this is the dominant segment by volume.
- Profile Summary: Mass-affluent traditional segment – lower credit limit, more bank visits and calls, low online usage. Likely older or less digitally inclined users. These are traditional customers that add less value and need more in person banking to gain their trust.

Cluster 1 :

- Avg_Credit_Limit: ₹1,41,040 → very high credit limit.
- Total_Credit_Cards: 8.74 → owns many credit cards, highly credit-savvy.
- Total_visits_bank: 0.6 → rarely visits branches.
- Total_visits_online: 10.9 → heavy digital user.
- Total_calls_made: 1.08 → low reliance on phone support.
- Count: 50 customers → a small but elite group.
- Profile Summary: Digitally savvy premium segment – high credit exposure, highly engaged online, minimal branch/call use. Likely affluent, younger, and tech-forward customers. This is the premium segment that any bank should handle with care. They must be given more premium digital offerings, loyalty programs, or concierge services, promotions, technical upgrades and solutions to retain them and attract more business with technology and least human intervention.

Now we will check k=3 to understand if it gives better business sense and also to compare with the 3 clusters we received from k-means clustering.

Cluster Profiling (k=3)

HC_Clusters	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_Eeach_segment
0	33713.178295	5.511628	3.485788	0.984496	2.005168	387
1	141040.000000	8.740000	0.600000	10.900000	1.080000	50
2	12197.309417	2.403587	0.928251	3.560538	6.883408	223

Table 14: HC_Clusters Profile with k=3

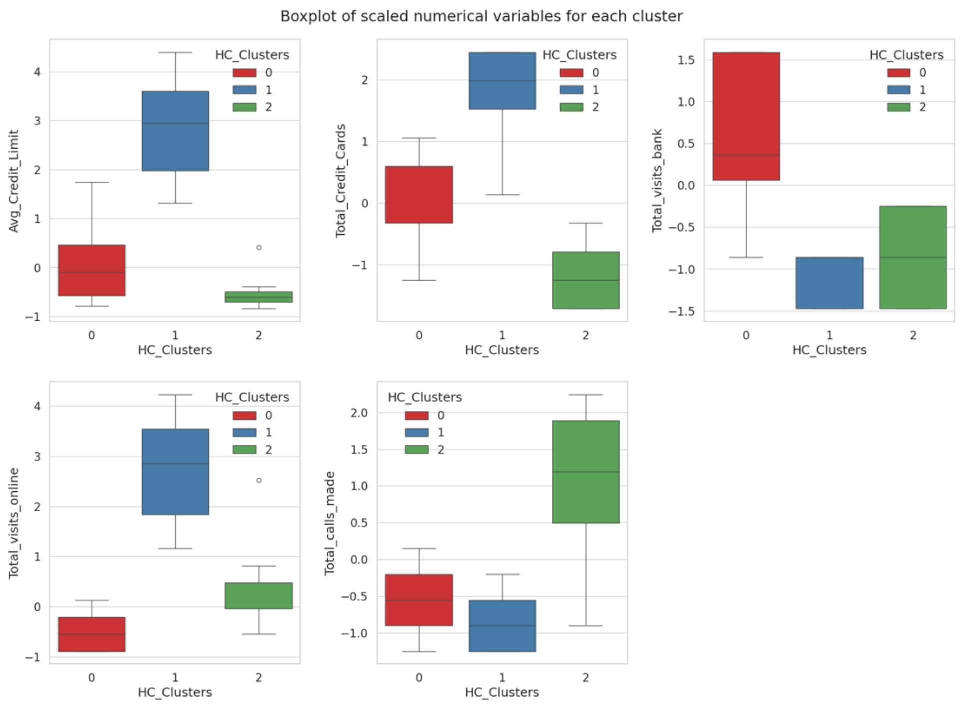

Figure 20: Boxplot of scaled numerical variables for each cluster (Hierarchical Clustering and k=3)

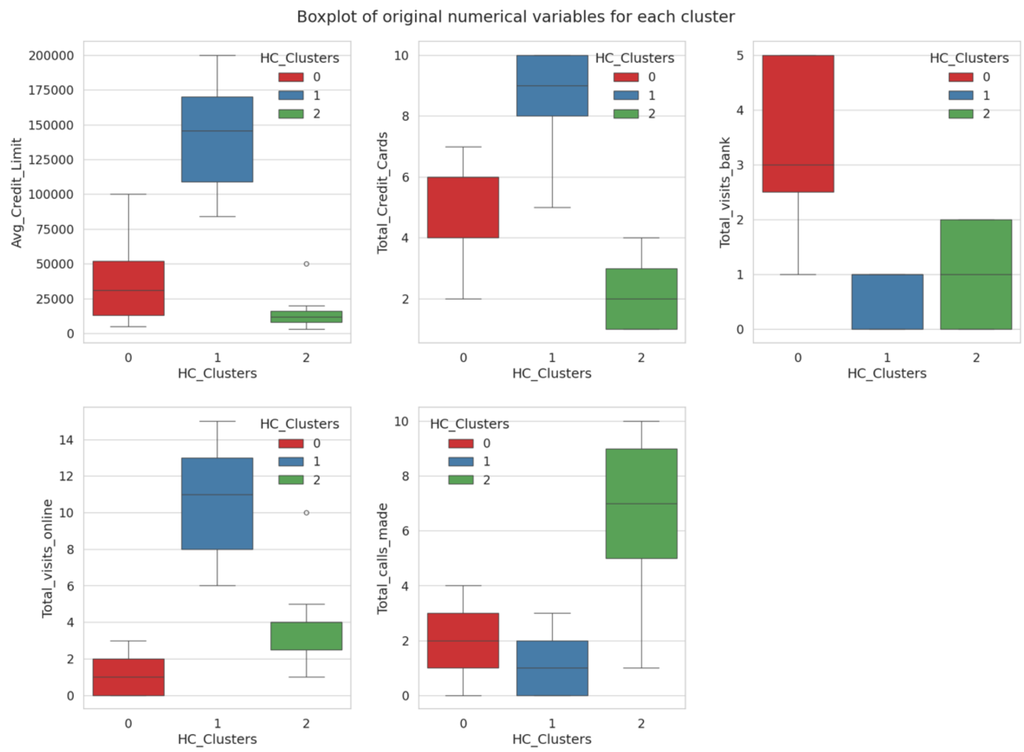

Figure 21: Boxplot of original numerical variables for each cluster (Hierarchical Clustering and k=3)

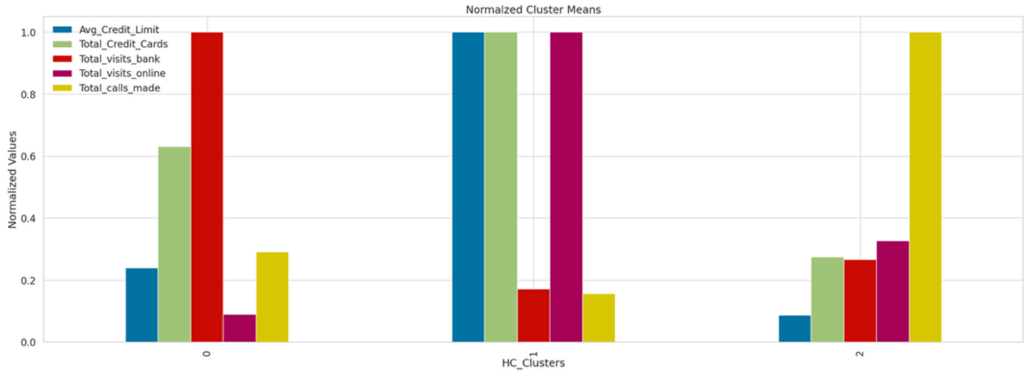

Figure 22: Normalised Mean Values vs HC_Clusters

Insights

Cluster 0: Mass-Affluent Traditional Segment

- Avg_Credit_Limit: ₹33,713 → Moderate credit exposure
- Total_Credit_Cards: ~5.5 → Above-average number of cards
- Total_visits_bank: 3.49 → Highest branch engagement
- Total_visits_online: 0.98 → Very low online usage
- Total_calls_made: ~2 → Moderate use of phone banking
- Size: 387 customers (majority cluster)
- **Insight:** These customers prefer physical banking (branch) and are less comfortable with digital channels. Likely older, conservative users with decent credit access.
- **Recommendations:** Company should promote assisted digital onboarding, simplify branch-to-digital transitions, offer loyalty rewards for branch visits.

Cluster 1: Premium Digital Power Users

- Avg_Credit_Limit: ₹1,41,040 → Exceptionally high credit limits
- Total_Credit_Cards: 8.74 → Very high product penetration
- Total_visits_bank: 0.6 → Least likely to visit branches
- Total_visits_online: 10.9 → Extremely high online activity
- Total_calls_made: ~1 → Minimal support needs
- Size: 50 customers (smallest, elite segment)
- **Insight**: These are wealthy, digitally native customers. They prefer self-service channels, may be more urban and younger, and expect seamless digital experiences.
- **Recommendations**: Offer premium digital products, priority servicing, and curated offers. Maintain strong app/UI innovation.

Cluster 2: Low-Income, High-Support Segment

- Avg_Credit_Limit: ₹12,197 → Lowest credit limit
- Total_Credit_Cards: 2.4 → Low product ownership

- Total_visits_bank: 0.93 → Low physical interaction
- Total_visits_online: 3.56 → Moderate online usage
- Total_calls_made: 6.88 → Very high support dependency
- Size: 223 customers
- **Insight:** These customers are financially underserved and rely heavily on phone support, possibly due to issues or low digital proficiency. Likely lower-income, rural or semi-urban customers.
- **Recommendations:** Focus on financial literacy, reduce call center load via proactive communication, offer basic digital banking tools.

K-means (KM) vs Hierarchical Clustering (HC)

A. **From Cluster Profiling, we see that**:

- At k=3, both techniques give clusters of the following types :

1. Mass-Affluent Traditional Segment
2. Premium Digital Power Users
3. Low-Income, High-Support Segment The count is same for premium segment and at +- 1 for the other 2. Thus , clusters are highly comparable in nature from the 2 techniques

B. **Using Silhouette Scores** is more time-consuming with Dendrograms for HC and gives k=2 for Euclidean Average (Best Cophenetic Correlation of 0.9) HC in contrast to k=3 for K-means Clustering which also makes better business sense.

C. At k = 3 clusters, K-means clustering takes more time for execution as compared to that of HC, which is highly efficient, but Dendrogram formation is the most time-consuming as mentioned above.

D. Refer business report for detailed comparison of statistics as well as Silhouette scores for each technique. However, At k=3, Silhouette Score for HC = 0.515 (Euclidean Distance Average Linkage) and Silhouette Score for K-Means Clustering = 0.515. Thus, scores are same at 3 decimal places.

E. Adjusted Rand Index (ARI) - Measures Agreement between Clusterings

ARI= 0.994

- The score ranges from -1 to 1:

a. 1.0 = Perfect agreement between two clustering labels.

b. 0.0 = Random labeling (no agreement beyond chance).

c. < 0.0 = Worse than random (very rare).

Our value of 0.994 is very close to 1, so:

The two clustering techniques being compared are almost identical.

F. **Add-on: PCA for Visualization** Let's use PCA to reduce the data to two dimensions and visualize it to see how well-separated the clusters are for both the Methods used

The first two principal components explain 83.16% of the variance in the data.

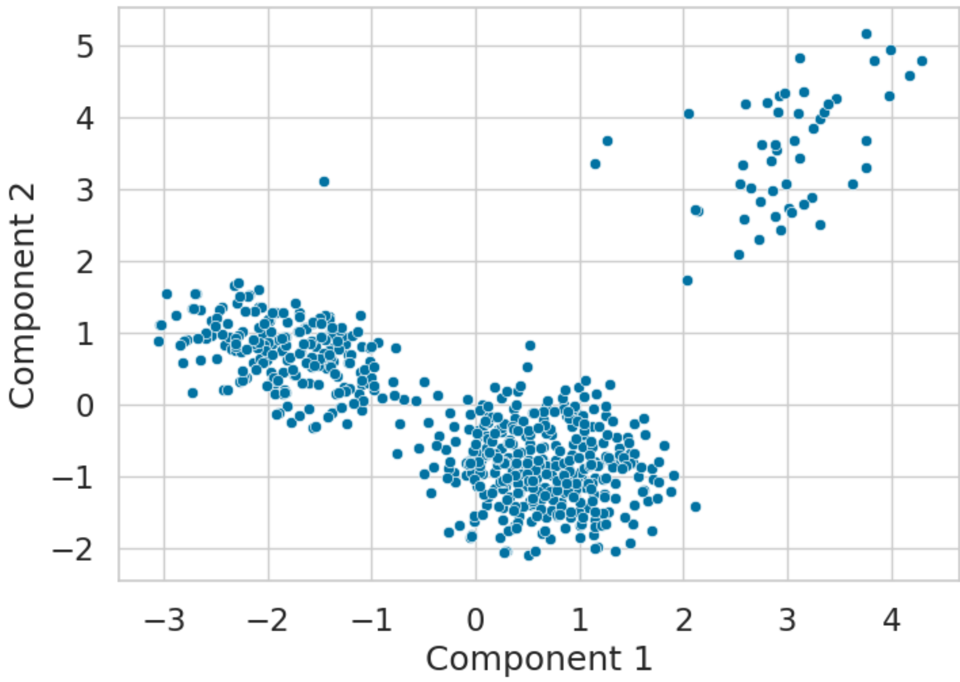

Figure 23: Scatter Plot of Reduced Data (PCA)

Figure 24: Comparison of Hierarchical & K-Means Clustering (PCA)

It is evident that except one observation being classified differently, rest all observations are classified same by the both techniques.

Actionable Insights & Recommendations

Actionable Insights:

- K-means and Hierarchical Clustering techniques were used to get 3 Discrete Clusters to segment customers based on different banking traits. Clusters formed are 99.4% in agreement with each other with same number of observations in Premium Segment (50) and just 1 observation that has moved from Low-Income, High-Support Segment to Mass-Affluent Traditional Segment in Hierarchical Clustering
- Silhouette Score of 0.515 and Cophenetic Correlation of 0.9 is obtained implying good quality clusters
- The segments obtained are as follows:

Cluster 0: Mass-Affluent Traditional Segment

- Avg_Credit_Limit: ₹33,713 → Moderate credit exposure
- Total_Credit_Cards: ~5.5 → Above-average number of cards
- Total_visits_bank: 3.49 → Highest branch engagement
- Total_visits_online: 0.98 → Very low online usage
- Total_calls_made: ~2 → Moderate use of phone banking
- Size: 387 customers (majority cluster)

- Insight: These customers prefer physical banking (branch) and are less comfortable with digital channels. Likely older, loyal, conservative users with decent credit access.
- Recommendations: Company should promote assisted digital onboarding, simplify branch-to-digital transitions, offer loyalty rewards for branch visits.

Cluster 1: Premium Digital Power Users

- Avg_Credit_Limit: ₹1,41,040 → Exceptionally high credit limits
- Total_Credit_Cards: 8.74 → Very high product penetration
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- Total_visits_online: 10.9 → Extremely high online activity
- Total_calls_made: ~1 → Minimal support needs
- Size: 50 customers (smallest, elite segment)
- Insight: These are wealthy, digitally native customers. They prefer self-service channels, may be more urban and younger, and expect seamless digital experiences.
- Recommendations: Offer premium digital products, priority servicing, and curated offers. Maintain strong app/UI innovation.

Cluster 2: Low-Income, High-Support Segment

- Avg_Credit_Limit: ₹12,197 → Lowest credit limit
- Total_Credit_Cards: 2.4 → Low product ownership
- Total_visits_bank: 0.93 → Low physical interaction
- Total_visits_online: 3.56 → Moderate online usage
- Total_calls_made: 6.88 → Very high support dependency
- Size: 223 customers
- Insight: These customers are financially underserved and rely heavily on phone support, possibly due to issues or low digital proficiency. Likely lower-income, rural or semi-urban customers and need more frequent assistance.
- Recommendations: Focus on financial literacy, reduce call center load via proactive communication, offer basic digital banking tools.

Business Recommendations:

Company should adopt a segment-specific marketing and service strategy. Combine data-driven insights with operational efficiency to improve customer experience, increase loyalty, and drive revenue across all segments.

For Cluster 0 (Mass-Affluent Traditional Segment – 59% customer base):

This is traditional loyal segment and needs to be engaged using cross selling, up selling, etc - Promote assisted digital onboarding at branches. - Roll out in-branch loyalty rewards to retain engagement. - Use hybrid engagement (branch + digital nudges via SMS/email/app). - Segment for cross-sell—especially premium credit cards, wealth management.

For Cluster 1 (Premium Digital Users – 8% customer base):

These are high value premium tech savvy customers and need lucrative retention schemes , otherwise they may move elsewhere for better services - Provide priority service lanes, personal relationship managers, Digital-first promotions and curated offers (e.g., exclusive travel cards). - Focus on digital experience excellence—optimize app UI/UX, AI chatbots, instant query resolution. - Consider reward partnerships (luxury brands, travel, fintech platforms). - Treat them as brand ambassadors—targeted referral schemes and loyalty rewards.

For Cluster 2 (Low-Income, High-Support – 33% customer base):

This is where a lot of untapped potential lies because they are not using a lot of services and have issues. This segment needs improvement to avoid churning out. - Launch financial literacy programs to empower and educate. - Offer basic digital training to reduce dependency on phone support. - Introduce low-fee digital banking tools, simplified mobile app versions. - Proactive customer service to reduce repeat calls and enhance satisfaction.

Other takeaways :

- 67% of customers had ≤2 online visits/year — suggests low digital penetration. Thus, there is opportunity to boost mobile/online banking adoption across all but Cluster 1.
- High call volumes in Cluster 2 drives up operational costs. Reducing call dependency via digitization and targeted education can significantly optimize resources. Same with the traditional segment of Cluster 0 that visits bank often.